

Lesson 05

Divisibility

5.1 Introduction

From your school days you are very familiar with finding the quotient and remainder after dividing a positive integer by another positive integer. In that procedure first integer is called the dividend and the second integer is called the divisor. For an example if you divide 123 by 8 then the quotient is 15 and the remainder is 3. When the remainder is 0, we say that divisor divides dividend properly. We are learning the properties of this special situation now.

5.2 Divisibility

5.2.1 Definition

If d and n are integers, we say that the “integer d divides the integer n ” if there is an integer k such that $n = dk$.

Note:

Symbolically d divides n denote as $d|n$. The symbol $d \nmid n$ means that d does not divide n .

Alternative terms for d divides n are,

d is a divisor of n
 d is a factor of n
 n is a multiple of d
 n is divisible by d

4.2.2 Example

Since $24 = 8 \times 3$, $8|24$.

Since $24 = 3 \times 8$, $3|24$.

Since $24 = (-3) \times (-8)$, $(-3)|24$.

Since $(-24) = 3 \times (-8)$, $3|(-24)$.

Since $(-24) = (-3) \times 8$, $(-3)|(-24)$.

Since $8 = 8 \times 1$, $8|8$.

Since $8 = 1 \times 8$, $1|8$.

Since $15 \neq 7 \times k$ for any integer k , $7 \nmid 15$.

5.2.3 Theorem (Divisibility Properties)

For any integers n , m and d ,

- (a) $d|0$
- (b) If $0|n$ then $n = 0$
- (c) $1|n$
- (d) $n|n$
- (e) If $n|1$, then $n = 1$ or $n = -1$.
- (f) If $d|n$ and $n|m$, then $d|m$.
- (g) If $d|n$ then for every integer a , $ad|an$.
- (h) If for non zero integer a , $ad|an$, then $d|n$.
- (i) If $d|n$ and $d|m$, then for any integers a, b $d|an + bm$.
- (j) If d and n are positive and $d|n$, then $d \leq n$.

Proof:

(a) Since $0 = d \cdot 0$, $d|0$.

(b) Let $0|n$. Then $n = 0 \cdot k$ for some integer k
 $\therefore n = 0$.

(c) Since $n = 1 \cdot n$, $1|n$.

(d) Since $n = n \cdot 1$, $n|n$.

(e) Let $n|1$. Then, $1 = nk$ for some integer k .

Since 1 and -1 are only factors of 1, n must be 1 or -1.

(f) Suppose $d|n$ and $n|m$

Then, $n = dk_1$ and $m = nk_2$ for some integers k_1, k_2

$\therefore m = (dk_1)k_2 = dk$, where $k = k_1k_2$ is an integer

$\therefore d|m$

(g) Suppose $d|n$

Then, $n = dk$ for some integer k .

$\therefore an = ad \cdot k$, where k is an integer

$\therefore ad|an$

(h) Suppose $ad|an$ and $a \neq 0$,

Then $an = ad \cdot k$ for some integer k .

$\therefore n = dk$ (Since $a \neq 0$)

$\therefore d|n$

(i) Suppose $d|n$ and $d|m$

Then, $n = dk_1$ and $m = dk_2$ for some integers k_1, k_2

$\therefore an + bm = a(dk_1) + b(dk_2) = d(ak_1 + bk_2) = dk$

where $k = ak_1 + bk_2$ is an integer

$\therefore d|an + bm$

(j) Suppose d and n are positive and $d|n$.

Then $n = dk$ for some integer k .

Since n and d are positive, k must be positive.

Assume that $d > n$. Then $kd > kn$

$\therefore n > kn \quad \therefore n(1-k) > 0$. Since $n > 0$, $1-k > 0$

Hence $k < 1$. This is a contradiction.

Thus $d \leq n$

5.2.4 Lemma

(a) If $d|a$ then $-d|a$ and $d|-a$

(b) If $d|a$ then $d||a|$

Note: $|a|$ equals a if $a \geq 0$ and equals $-a$ if $a < 0$.

Proof:

(a) Suppose $d|a$

Then $a = dk$ for some integer k .

$\therefore a = (-d)(-k)$

Since $-d$ and $-k$ are also integers, $-d|a$

Also since $-a = d(-k)$, $d|(-a)$.

(b) Suppose $d|a$

Assume that a is non negative

Then $|a| = a$. Since $d|a$, we get $d||a|$.

Now suppose a is negative.

Then $|a| = -a$. Since $d|a$, $d|-a$

$$\therefore d \parallel a.$$

$$\text{Hence } d \parallel a$$

5.2.5 Examples

State whether true or false. Justify your answer.

(i) If $a|b$ and $b|a$ then $a = b$.

(ii) If $a|b$ then $(-a)|b$

(iii) If $a|bc$ then $a|b$ or $a|c$

Answers:-

(i) This is false. Observe that $3|(-3)$ and $(-3)|3$ but $3 \neq -3$. Hence the statement is false.

(ii) This statement is true. Let $a|b$, then there exist an integer k such that $b = ka$.

This can be written as $b = (-k)(-a)$. Hence by definition of divisibility $(-a)|b$.

(iii) False. $6|2 \cdot 9$ but $6 \nmid 2$ and $6 \nmid 9$

Activity

State whether true or false. Justify your answer

(i) If $a|b$ and $b|a$ then $a = b$ or $b = -a$

(ii) If $a|(b + c)$ then $a|b$ or $a|c$

(iii) If $a \nmid b$ then $b \nmid a$

(iv) If $a|b$ and $c|d$ then $ac|bd$

(v) If $a^3|b^3$ then $a|b$

(vi) If $a^2|b^3$ then $a|b$

?

Prove or disprove : If $a^3|b^2$ then $a|b$.

Following examples illustrates how to use mathematical induction to verify some non trivial divisibility identities in integers.

5.2.6 Examples

Using the Principle of Mathematical Induction, for each $n \in \mathbb{N}$, verify the following identities.

(i) $6|2n^3 + 3n^2 + n$

(ii) $9|2^{4n} + 3n - 1$

(iii) $8|5^{2n} + 7$

Answers:-

(i) Let $P(n)$ be the statement $6|2n^3 + 3n^2 + n$. (or $2n^3 + 3n^2 + n$ is divisible by 6)

When $n = 1$, $2n^3 + 3n^2 + n = 2(1)^3 + 3(1)^2 + (1) = 6$. But $6|6$. Hence $P(1)$ is true.

Let k be an arbitrary positive integer. We would like to show that $P(k)$ implies $P(k + 1)$.

Assume $P(k)$ is true; that is $6|2p^3 + 3p^2 + p$. Now;

$$\begin{aligned} 2(p + 1)^3 + 3(p + 1)^2 + (p + 1) &= 2(p^3 + 3p^2 + 3p + 1) + 3(p^2 + 2p + 1) + p + 1 \\ &= (2p^3 + 3p^2 + p) + 6(p^2 + 2p + 1) \end{aligned}$$

But by assumption $6|(2p^3 + 3p^2 + p)$, also $6|6(p^2 + p + 1)$

Therefore $6|2(p + 1)^3 + 3(p + 1)^2 + (p + 1)$.

So, if $P(k)$ is true, then $P(k + 1)$ is true. Hence by Principle of Mathematical Induction the statement is true for every $n \in \mathbb{N}$.

(ii) Let $P(n)$ be the statement $9|2^{4n} + 3n - 1$. When $n = 1$, $2^{4n} + 3n - 1 = 2^4 + 3 - 1 = 18$.

Observe that $9|18$. Hence $P(1)$ is true.

Now assume $P(k)$ is true, that is $9|2^{4k} + 3k - 1$.

Observe that $2^{4(k+1)} + 3(k + 1) - 1 = 2^{4k}2^4 + 3k + 2 = 16(2^{4k} + 3k - 1) - 45k + 18$

But $-45k + 18 = 9(-5k + 2)$. This implies $9|(-45k + 18)$.

By assumption $9|2^{4k} + 3k - 1$. Hence $9|2^{4(k+1)} + 3(k+1) - 1$.

So, if $P(k)$ is true, then $P(k+1)$ is true. Hence by Principle of Mathematical Induction the statement is true for every $n \in \mathbb{N}$.

(iii) Let $P(n)$ be the statement $8|5^{2n} + 7$. When $n = 1$, $5^{2 \cdot 1} + 7 = 32$.

Observe that $8|32$. Hence when $n = 1$, the statement is true.

Assume $P(k)$ is true, that is $8|5^{2k} + 7$.

Observe that $5^{2(k+1)} + 7 = 25 \cdot 5^{2k} + 7 = 25(5^{2k} + 7) - 168$.

Now by assumption $8|5^{2k} + 7$ and also it is clear $8|(-168)$

Therefore $8|25(5^{2k} + 7) - 168$, means $8|5^{2(k+1)} + 7$.

So, if $P(k)$ is true, then $P(k+1)$ is true. Hence by Principle of Mathematical Induction the statement is true for every $n \in \mathbb{N}$.

Note:-Did you observe, throughout above three examples, we use the divisibility property,

“If $d|n$ and $d|m$, then for any integers a, b $d|an + bm$ ”.

Activity

Using the Principle of Mathematical Induction, for each $n \in \mathbb{N}$, verify the following identities.

(i) $3|n(2n^2 + 7)$

(ii) $15|2^{4n} - 1$

(iii) $21|4^{n+1} + 5^{2n-1}$

?

Let a, b, c, d, e are integers. Suppose $1|a$, $2|ab$, $3|abc$, $4|abcd$, and $5|abcde$

Is that necessary to be $1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 | abcde$?

5.3 Even and Odd integers

You all are well familiar with even and odd integers. However here we give a definition of even and odd integers based on the divisibility.

5.3.1 Definition-Even and Odd integers

An integer n is called Even if $2|n$ and n is called Odd if $2 \nmid n$.

Note: This definition is coincided with more familiar definition of Even and Odd integers;

An integer n is Even if $n = 2k$ for some integer k

An integer n is Odd if $n = 2k - 1$ for some integer k .

Following examples and activities are related to the concept of even and odd integers.

5.3.2 Examples

Prove the following statements.

- (i) The sum of any two odd integers is even.
- (ii) If the square of an integer is odd, then that integer must be odd.
- (iii) If the product of two integers is even, then at least one of them must be even.

Answers:-

For (i), you can give a direct proof, but to prove (ii) and (iii) either you must prove the contra positive of each statements or must use the contradiction method. Recall that contra positive of the statement $p \Rightarrow q$ is $\text{not } q \Rightarrow \text{not } p$.

(i) Let m, n be odd integers. Then $m = 2i - 1$ and $n = 2j - 1$ for some $i, j \in \mathbb{N}$.

Hence $m + n = 2i - 1 + 2j - 1 = 2(i + j - 1)$, which implies $m + n$ is even.

(ii) The contraposition of the statement “If the square of an integer is odd, then that integer must be odd” is “If the integer is even then the square of that integer is even”. Now we can prove the contraposition; Any even integer is of the form $2k$. Now the square of that is

$4k^2 = 2 \cdot 2k^2$, which is even. So we prove the contraposition. But contraposition of a statement is equivalent to that statement. Hence we prove the statement.

(iii) The contraposition of the statement “If the product of two integers is even, then at least one of them must be even.” is “If both integers are odd then the product of those two integers is odd”. Now let’s prove the contraposition; Let a, b are odd integers. Therefore $a = 2m - 1$ and $b = 2n - 1$. Hence $ab = (2m - 1)(2n - 1) = 2(2mn - (m + n)) + 1$. This proves ab is even.

Review Questions

(1) Let m, n are positive integers such that $m|n$. Prove each of the followings.

(i) $m \leq n$

(ii) $\frac{n}{m} \leq n$

(ii) $\frac{n}{m} | n$

(2) Let n be positive integer. Show that if $n = kl$ with $1 \leq k \leq l < n$, then $k \leq \sqrt{n}$.

(3) Using the Principle of Mathematical Induction prove each of the followings.

(i) $n^4 + 2n^3 + n^2$ is divisible by 4.

(ii) $4^{2n} + 10n - 1$ is divisible by 25.

(iii) $n^5 - n$ is divisible by 30.

(4) Prove each of the followings.

(i) The sum of any two even integers is even.

(ii) The sum of any two odd integers is even.

(iii) The product of any two consecutive integers is even.

(iv) If the square of an integer is even then that integer must be even.

(v) If the product of two integers is odd, then both must be odd.

(5) State whether true or false. Justify your answer.

- (i) Zero is neither even nor odd.
- (ii) If the sum of two integers is odd, then exactly one must be odd.
- (iii) If the sum of three integers is odd, then exactly one must be odd.
- (iv) If the sum of three integers is odd then at least one must be odd.
- (v) If the product of three integers is odd, then all three are odd.
- (vi) If the product of three integers is even, then all three are even.